$$= \frac{(1216)a}{16}\left[1 - \frac{1}{(1.0133)^{180}}\right]$$

$$= \frac{1216}{16}a\left[1 - \frac{1}{9.772}\right]$$

$$= \frac{1216}{16}a\left[\frac{9.772 - 1}{9.772}\right]$$

$$= \frac{1216}{16}a\left[\frac{8.772}{9.772}\right]$$

$$\Rightarrow a = \frac{7,00,000 \times 16 \times 9772}{1216 \times 8772} \approx ₹\ 10,261$$

Example 7.8

The chairman of a society wishes to award a gold medal to a student getting highest marks in Business Mathematics and Statistics. If this medal costs to ₹ 9,000 every year and the rate of compound interest is 15%, then what amount is to be deposited now.

Solution:

Here a = 9,000 and i = 0.15

$$P = \frac{a}{i}$$

$$= \frac{9000}{0.15}$$

$$= \frac{9,00,000}{15}$$

= 60,000.

Therefore the amount to be deposited is ₹ 60,000.

Example 7.9

A limited company wants to create a fund to help their employees in critical circumstances. The estimated expenses per month is ₹18,000. Find the amount to be deposited by the company if the rate of compound interest is 15%.

Solution:

Here a = 18,000 ; i = 0.15 and k = 12.

$$P = \frac{a}{i/k} = \frac{18,000}{0.15/12}$$

$$= \frac{18,000}{0.15} \times 12$$

$$= \frac{18,00,000 \times 12}{15}$$

= 14,40,000

Therefore the amount to be deposited is ₹14,40,000.

Example 7.10

Find the annual rate of interest, to get a perpetuity of ₹675 for every half yearly from the present value of ₹30,000.

Solution :

Here P=30,000 ; a = 675 ; k = 2, i = ?

$$P = \frac{a}{i/k}$$

$$30,000 = \frac{675}{\frac{i}{2}}$$

$$= \frac{1350}{i}$$

$$\Rightarrow i = \frac{1350}{30,000} = \frac{135}{3000} = 0.045$$

Rate of interest (i)= 0.045 × 100% = 4.5%

Exercise 7.1

1. Find the amount of an ordinary annuity of ₹3,200 per annum for 12 years at the rate of interest of 10% per year. $[(1.1)^{12} = 3.1384]$.
2. If the payment of ₹2,000 is made at the end of every quarter for 10 years at the rate of 8% per year, then find the amount of annuity. $[(1.02)^{40} = 2.2080]$.
3. Find the amount of an ordinary annuity of 12 monthly payments of ₹1, 500 that earns interest at 12% per annum compounded monthly. $[(1.01)^{12} = 1.1262]$.

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